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REVIEW

Combined Pulmonary Fibrosis and Emphysema: A Narrative Review

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Abstract

Combined pulmonary fibrosis and emphysema (CPFE) syndrome refers to co-occurrence of two disease processes in the lung that can be difficult to diagnose but is associated with high morbidity and mortality burden. Diagnosis of CPFE is challenging because the two diseases can counterbalance respective impairments resulting in deceivingly normal-appearing chest radiography and spirometry in a dyspneic patient. Although an international committee published the terminology and definitions of CPFE in 2022, consensus on exact diagnostic criteria and optimal management strategy is yet to be determined. Herein, we provide a narrative review summarizing the literature on CPFE from 1990 to 2022, including historical background, epidemiology, pathogenesis, clinical features, imaging and pulmonary function findings, diagnosis, prognosis, complications, and treatment. Although CPFE was initially conceived as a variant presentation of idiopathic pulmonary fibrosis, it has been recognized to occur in patients with a wide variety of interstitial lung diseases, including connective tissue disease–associated interstitial lung diseases, and hypersensitivity pneumonitis. The affected patients have a heightened risk for pulmonary hypertension and lung cancer. Clinicians need to recognize the characteristic presenting features of CPFE along with prognostic implications of this entity.

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Dyspnea is a common symptom for which patients seek medical advice. Most often, dyspnea originates from a respiratory or cardiovascular disorder. The diagnostic considerations and evaluation will differ depending, in part, on whether dyspnea is acute or chronic. In either case, clinical history and physical examination may be sufficient for diagnosis and subsequent management. Otherwise, additional diagnostic evaluation including blood tests, chest radiography, electrocardiography, spirometry, and other imaging studies may be needed. In particular, spirometry is used to screen for evidence of pulmonary impairment.

Chronic obstructive pulmonary disease (COPD), emphysema, and interstitial lung diseases (ILDs) are among the most common chronic respiratory disorders. Chest radiography and spirometry usually demonstrate abnormalities in patients with chronic dyspnea due to lung disease. However, it is increasingly recognized that two chronic respiratory disorders, emphysema and ILD, can co-occur in some patients, confounding physiological measurements, and leading to missed diagnoses. This phenomenon is referred to as combined pulmonary fibrosis and emphysema (CPFE). Accurately speaking, as the name implies, CPFE is a clinico-physiologic-radiologic syndrome usually characterized by the presence of emphysema (predominantly in the upper lungs, not necessarily meeting COPD criteria) and fibrosis (predominantly in the lower lungs) (Table 1).[1] Some affected patients may manifest airflow limitation on pulmonary function test (PFT) characteristic of COPD, although many do not. While this entity was initially described in the context of idiopathic pulmonary fibrosis (IPF) developing in patients with pre-existing emphysema, similar phenomena have been observed with a widening array of ILDs, including non-IPF interstitial pneumonias,



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ARTICLE HIGHLIGHTS

- Combined pulmonary fibrosis and emphysema (CPFE) is a clinicoradiologic syndrome characterized by chronic respiratory symptoms, radiologic evidence of both parenchymal fibrosis and emphysema, and gas exchange abnormalities.
- In the evaluation of a dyspneic patient, the abnormalities of parenchymal fibrosis and emphysema may be difficult to recognize because the counterbalancing effects of these two entities may obscure their combined presence as assessed by chest radiography and spirometry.
- High-resolution computed tomography of the chest and measurement of diffusing capacity are essential in the diagnostic evaluation for CPFE.
- CPFE syndrome is being recognized with a broadening spectrum of interstitial lung diseases, and management strategies depend, in part, on the specific nature of the underlying interstitial lung disease.

connective tissue disease (CTD)—related ILDs, fibrotic (chronic) hypersensitivity pneumonitis, and others.[2-4]

Emphysema and fibrosis are common sequelae of lung injury, and smoking can lead to both. Thus, it is not surprising that both entities can be encountered in smokers, sometimes simultaneously. Although CPFE has become recognized as a form of smoking-related lung disease, it has not yet been included in the reports on cigarette smoking by the World Health Organization or the Centers for Disease Control and Prevention in the United States.[2,5] An increasing number of respiratory-related deaths has been reported worldwide, [6] including more than 3 million COPD-related deaths per year.[2,7] Our understanding of the pathophysiology of COPD, emphysema, and pulmonary fibrosis has advanced over the years, leading to the development of novel therapies that can improve survival and quality of life. As such, it is important for clinicians to recognize the clinico-physiologic-radiologic presentation associated with CPFE and be able to diagnose this serious disorder.

The entity of CPFE is underrecognized in clinical practice because the abnormalities associated with parenchymal fibrosis and emphysema can be obscured by their combined presence in the evaluation of a dyspneic patient. The loss of elastic recoil of the lung (or increased compliance) in emphysema is counteracted by the increased recoil (or reduced compliance) in fibrosis. This canceling effect can result in perfectly normal spirometry and lung volumes if the two effects are evenly matched. Even if not matched, the lesser effect mitigates the apparent abnormality related to the dominant process. However, both processes reduce diffusion capacity of the lungs for carbon monoxide (DLCO), which is often severely reduced in such cases. An international committee, constituted of American Thoracic Society,

TABLE 1. Definitions for Combined Pulmonary Fibrosis and Emphysema and Related Terms

Term	Definition
Chronic obstructive pulmonary disease	Lung condition characterized by airflow limitation (indicated by low ratio of forced expiratory volume in the first second to forced vital capacity on spirometry).
Emphysema	Abnormal and permanent enlargement of airspaces distal to the terminal bronchiole associated with loss of airspace walls.
Pulmonary fibrosis	Parenchymal fibrosis associated with architectural distortion of lung tissue.
Combined pulmonary fibrosis and emphysema	Combined presence of parenchymal fibrosis and emphysema (often without airflow limitation).
Interstitial lung abnormalities	Incidentally detected parenchymal abnormalities on chest computed tomography scan suggestive of early interstitial lung disease (includes ground-glass or reticular opacities, architectural distortion, traction bronchiectasis/bronchiolectasis, honeycombing, and nonemphysematous cysts).

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European Respiratory Society, Japanese Respiratory Society, and The Latin American Thoracic Association, recently published a research statement on CPFE.[8] The statement contained the proposed definition of CPFE as well as a description of clinical manifestation and radiologic-histopathological characteristics. However, an optimal definition for CPFE requires further data accumulation.

We herein provide an updated review of CPFE in addition to the CPFE research statement[8] to cover clinical information including epidemiology discussed by subtypes of pulmonary fibrosis and pulmonary function, the key imaging findings for CPFE, as well as the background, diagnosis, treatment, prognosis, complications, and future directions.

METHODS

This review used online research databases, including MEDLINE, PubMed, and Google Scholar, to gather information on the definition, diagnosis, and treatment of CPFE. Articles dating from 1990 to August 15, 2022, were surveyed, particularly those that mentioned CPFE specifically or its constituent parts combined. Our keywords included emphysema, pulmonary fibrosis, smoking, ILDs, COPD, and combinations thereof. We included studies addressing CPFE specifically or tangentially when they referred to the aforementioned keywords in an epidemiological context.

HISTORICAL BACKGROUND

Although the combined presence of pulmonary fibrosis and emphysema on chest radiography or autopsy had been described in isolated cases in earlier years, in 1990, Wiggins et al[9] reported eight heavy smokers who manifested severe breathlessness and an atypical pattern of pulmonary function impairment, that is, severely reduced DLCO with well-preserved vital capacity and forced expiratory volume in 1 second (FEV_1). Upper-zone emphysema and lower-zone ILD ("fibrosing alveolitis") were observed on chest computed tomography (CT) images but difficult to appreciate on chest radiography. In 1991, Schwartz et al[10] investigated the effect of cigarette smoking in patients with IPF. Their data showed a strong correlation between pack-years and lower DLCO and higher lung volumes among patients with IPF.[10] Cottin et al[11] further studied the characteristics of CT-diagnosed CPFE, including not only IPF but other underlying ILD phenotypes, and found that all were current or prior smokers and that the presence of pulmonary hypertension is a critical prognostic factor regardless of ILD phenotype. In the following years, CPFE was incorporated into the classification of "smoking-related ILDs."[12]

EPIDEMIOLOGY

Data on the incidence of CPFE are limited because of underrecognition of this entity. Moreover, in the absence of a consensus definition, various diagnostic criteria have been used in the literature, further hindering the determination of incidence and prevalence. In analyzing chest CT findings in several prospective cohort studies, including the Framingham Heart Study and COPDGene (Examining the Genetic Factors That May Cause Chronic Obstructive Pulmonary Disease [COPD]) study, 4% to 9% of patients with COPD have been estimated to have interstitial lung abnormalities (ILAs) (ie, subclinical parenchymal opacities detected on chest CT) suggestive of early or mild pulmonary fibrosis.[13] Lee et al[14] recently reported the presence of ILAs in 28.4% of 363 patients with COPD in a longitudinal retrospective study; 3.6% manifested "fibrotic" ILAs.[14] In these studies, patients with a prior diagnosis of ILD were excluded; therefore, it is likely that the actual incidence of CPFE is much higher than these estimates. Such conjecture is supported by the histopathologic observations by Katzenstein[15], who described "smoking-related interstitial fibrosis" to be commonly present on lung specimens obtained from smokers undergoing lung tumor resection with no previous diagnosis of ILD; the majority of these patients manifested coexistent emphysema microscopically.[15] However, the true

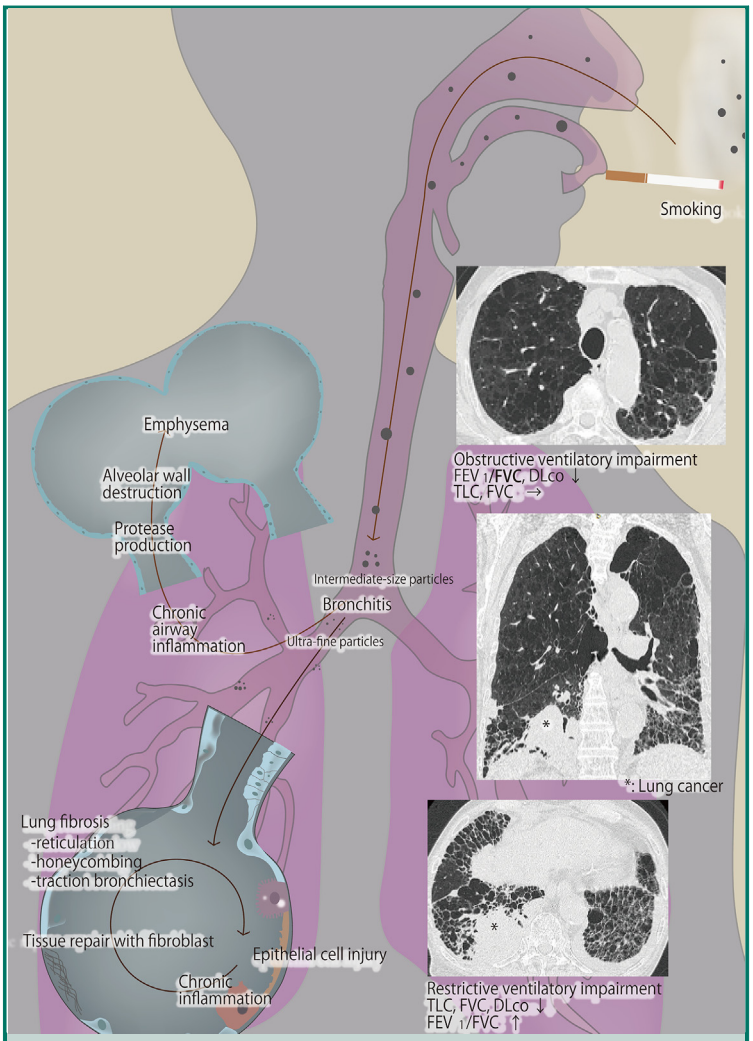


FIGURE 1. Pathogenesis of combined pulmonary fibrosis and emphysema. Tobacco smoke exposure can cause emphysema and fibrosis by different pathways. Chronic airway inflammation due to tobacco smoke causes thickening of airway walls and reduction in number of small airways. Ultrafine smoke particles can cause alveolar epithelial injury and chronic inflammation. This inflammation is associated with protease-antiprotease imbalance, oxidative stress, and enhanced apoptosis, leading to alveolar wall destruction (emphysema, computed tomography image on upper right). In contrast, the parenchymal repair process may result in inflammation and fibrosis, indicated by reticular and ground-glass opacities as well as honeycombing, architectural distortion, and traction bronchiectasis (lower right panel). DLCO = diffusing capacity for carbon monoxide; FEV_1 = forced expiratory volume in one second; FVC = forced vital capacity; TLC = total lung capacity.

relevance of this histopathologic entity to CPFE remains unclear because there are little data on correlating clinical, physiologic, and radiologic features.

Analysis of data in prior population-based studies on ILD also suggests a relatively high prevalence of CPFE. For example, in a study based on the population of Olmsted County, Minnesota, USA, 28% of patients with IPF also had a diagnosis of COPD (identified from the problem list on medical records).[16] A Korean nationwide study showed that, of patients with ILDs, 38% and 12% were smokers and had COPD, respectively.[17] Over the past decade, several studies have reported CPFE occurring in subjects with CTDs. One systematic review article showed that 30% of patients with CTD-ILDs had CPFE.[18] A French multicenter study described CPFE in patients with CTD-ILDs, most commonly rheumatoid arthritis–related ILD[19]; among these patients, 50% to 80% had a history of smoking,[20-22] and up to 48% had emphysema.[23] CPFE has also been described in other autoimmune diseases including systemic sclerosis[24,25] and microscopic polyangiitis.[26]

Subjects with other ILDs, including pneumoconiosis and sarcoidosis, have also been reported to manifest CPFE.[1] Of patients with pneumoconiosis, more than 80% are smokers and from 38% to 55% have emphysema.[27,28] Because 44% of patients with sarcoidosis are smokers according to a population-based study in Sapporo, Japan, these patients may also manifest the CPFE phenotype.[29]

The above-mentioned data show that CPFE is likely to be prevalent and often overlooked in patients with various forms of fibrotic ILDs. Although the smoking rate is gradually declining, the number of patients with COPD/emphysema and pulmonary fibrosis is increasing due to the aging population globally[1]; thus, the number of patients with CPFE is likely to increase as well.

PATHOGENESIS

As already noted, CPFE has been included under the rubric of smoking-related ILDs. However, CPFE is not a single distinct entity but rather a syndrome. Both emphysema[1] and a significant portion of cases of fibrosing

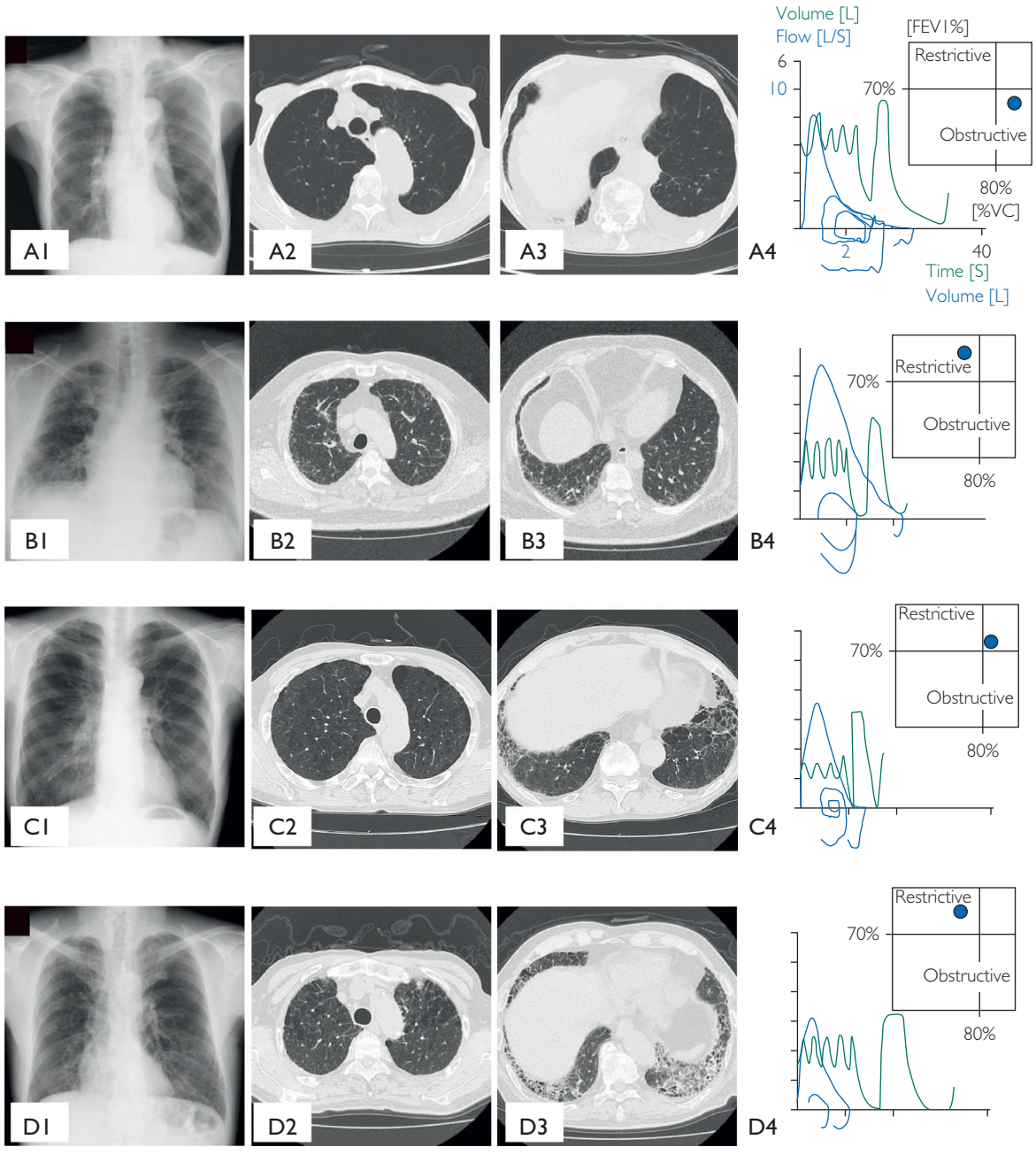


FIGURE 2. Chest computed (CT) tomography findings and corresponding pulmonary function test results. Radiologic images of patients with chronic obstructive pulmonary disease (A), idiopathic pulmonary fibrosis (IPF, B), and combined pulmonary fibrosis and emphysema (C, D). Frontal chest radiograph (A1 to D1), upper lung CT (A2 to D2), and lower lung CT (A3 to D3) with corresponding flow-volume and volume-time curve of each patient (A4 to D4). FEV_1 = forced expiratory volume in one second; VC = vital capacity.

ILDs[30] are strongly correlated with cigarette smoking, male sex, and older age. Cigarette smoking causes lung inflammation, which leads to pathways of parenchymal destruction and repair associated with emphysema and pulmonary fibrosis (Figure 1). Thus, in

some patients with CPFE, exposure to tobacco smoke may lead to both emphysema and fibrosis. In other patients, the superimposed parenchymal fibrosis may be related to a separate disease process (eg, CTD).

Several studies have reported the role of certain genetic factors, such as mutations in peptidase D (*PEPD*),[31] telomerase reverse transcriptase (*TERT*),[32] surfactant protein C (*SFTPC*),[33] and adenosine triphosphate–binding cassette subfamily A member 3 (*ABCA3*),[34] in the development of CPFE. There are members within the same family who develop only emphysema or pulmonary fibrosis, whereas CPFE may be observed in others, particularly young nonsmokers and women harboring these mutations. In terms of telomere length, which is thought to be a marker of cell senescence with a potential pathogenetic role in both pulmonary fibrosis and emphysema, a large Mendelian randomization study showed a significant association between a short telomere and IPF, but not COPD.[35] Putting together the evidence to date, there appear to be diverse mechanisms involving both genetic and environmental factors for developing pulmonary fibrosis and emphysema as well as CPFE.

CLINICAL FEATURES

Patients with CPFE are more commonly men 60 to 80 years of age with smoking exposure who experience a gradual onset of dyspnea on exertion for several months to years, often accompanied by a chronic cough.[36,37] Oxygen desaturation with physical exertion is commonly present and is attributable to impaired gas exchange (reflected in low DLCO) related to the combined presence of pulmonary fibrosis and emphysema.[38] Although patients with advanced COPD may show Hoover's chest sign, laryngeal height shortness, a barrel-shaped chest, and purse-lip breathing,[39] those with CPFE typically do not. However, bibasilar fine inspiratory crackles (due to pulmonary fibrosis) are usually present while decreased breath sounds are noted over the upper lungs (due to emphysema), and digital clubbing may be observed.

PULMONARY FUNCTION TESTS

Emphysema is characterized by loss of alveolar tissue associated with loss of lung recoil (increased lung compliance) and increased volume (hyperinflation). In contrast, most ILDs result in fibroinflammatory parenchymal infiltration which increases lung recoil (or reduces lung compliance) and reduces lung volume, which is characteristic of restrictive lung impairment. The loss of lung recoil due to emphysema is counterbalanced by the increased recoil due to fibrosis. When these are evenly balanced, spirometry and lung volumes may be perfectly normal. Both emphysema and ILD, however, reduce the amount of alveolar-capillary membrane through which gas exchange occurs resulting in severely decreased DLCO and hypoxemia with exertion or even at rest. Isolated reduction in DLCO on PFT observed in patients with CPFE can also occur with pulmonary vascular disease (eg, pulmonary hypertension), anemia, hemoglobinopathies, and right-to-left shunt.

Pulmonary function testing shows varying patterns of abnormalities depending on the extent of emphysema vs fibrosis (Figure 2); the most consistent finding is reduced DLCO. The airway collapse, which occurs with the development of emphysema (loss of parenchymal elasticity), results in airflow slowing manifest as reduced FEV_1. However, reduced FEV_1 is improved when fibrotic changes with associated traction bronchiectasis are superimposed. Similarly, hyperinflation (reflected in increased total lung capacity) associated with emphysema is ameliorated with the restrictive process of ILD. Both emphysema and pulmonary fibrosis damage lung parenchyma and gas exchange function, thus decreasing DLCO through these two additive processes. Thus, a marked decrease in DLCO, which is out of proportion to measures of spirometry and lung volumes, characterizes CPFE; at times, the only pulmonary function abnormality in these patients is decreased

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TABLE 2. Subtypes of Fibrotic Interstitial Lung Diseases Associated With CPFE[a]

Disease category	Disease name
Idiopathic interstitial pneumonias	Idiopathic pulmonary fibrosis
	Idiopathic nonspecific interstitial pneumonia
CTD and vasculitis-associated ILD	Rheumatoid arthritis–associated ILD
	Systemic sclerosis–associated ILD
	Microscopic polyangiitis–associated ILD
Other	Fibrotic (chronic) hypersensitivity pneumonitis
	Pneumoconioses
	Sarcoidosis
	Unclassifiable interstitial lung disease
	Interstitial pneumonia with autoimmune features

[a]CPFE = combined pulmonary fibrosis and emphysema; CTD = connective tissue disease; ILD = interstitial lung disease.

DLCO.[40] For example, in a study of 179 subjects with CPFE (defined by ≥10% total lung involvement with fibrosis and emphysema, respectively), 35% manifested low DLCO with normal spirometry, whereas 34% and 25% had restrictive and obstructive patterns of impairment, respectively; the remaining 6% manifested a nonspecific pattern of abnormalities or normal results.[26]

A composite physiologic index, calculated using forced vital capacity, DLCO, and FEV_1, has been reported to account for coexisting emphysema in the staging of patients with IPF.[41] This index was reported as one of the best prognostic indicators for patients with fibrotic ILDs.[42,43] However, a separate single-center retrospective study that included a wide variety of patients with CPFE failed to confirm the prognostic value of this index.[44]

CHEST IMAGING

Because of limited spatial resolution, emphysematous changes are challenging to identify on chest radiography, especially in the early stages of disease. Smoking-related emphysema more prominently affects the upper lungs. The apparent reduction in vessel number and caliber and diaphragm flattening due to hyperinflation are the key indirect radiographic signs as emphysema worsens.[45] On the other hand, fibrotic changes manifest as reticular or ground-glass opacities predominantly in the lower lungs, with decreasing lung volumes as fibrosis increases. Thus, the hyperinflation associated with emphysema is canceled as the lung volume is reduced by parenchymal fibrosis on chest radiography.

Chest CT in patients with CPFE may show two types of emphysema in the upper lungs: paraseptal and centrilobular emphysema (Figure 2). Fibrotic ILD on chest CT is characterized by reticulation, traction bronchiectasis, architectural distortion, and/or ground-glass opacities and honeycombing. Emphysema and fibrosis are well distinguished on high-resolution chest CT (HRCT). In consideration of the importance of HRCT scans in the evaluation of patients with CPFE as well as other parenchymal lung diseases, the American College of Radiology recommends single breath-hold axial or helical acquisition using a reconstructed image thickness of 1.5 mm or less and moderately high spatial-frequency algorithm, such as a bone algorithm for lung images.[46]

DIAGNOSIS

The recognition of CPFE is relatively straightforward: the combined presence of emphysema (usually predominant in the upper lungs) and parenchymal fibrosis (usually predominant in the lower lungs) (Figure 2) is best depicted by HRCT. However, the diagnostic criteria have varied in studies published to date.[8] In some studies, CPFE

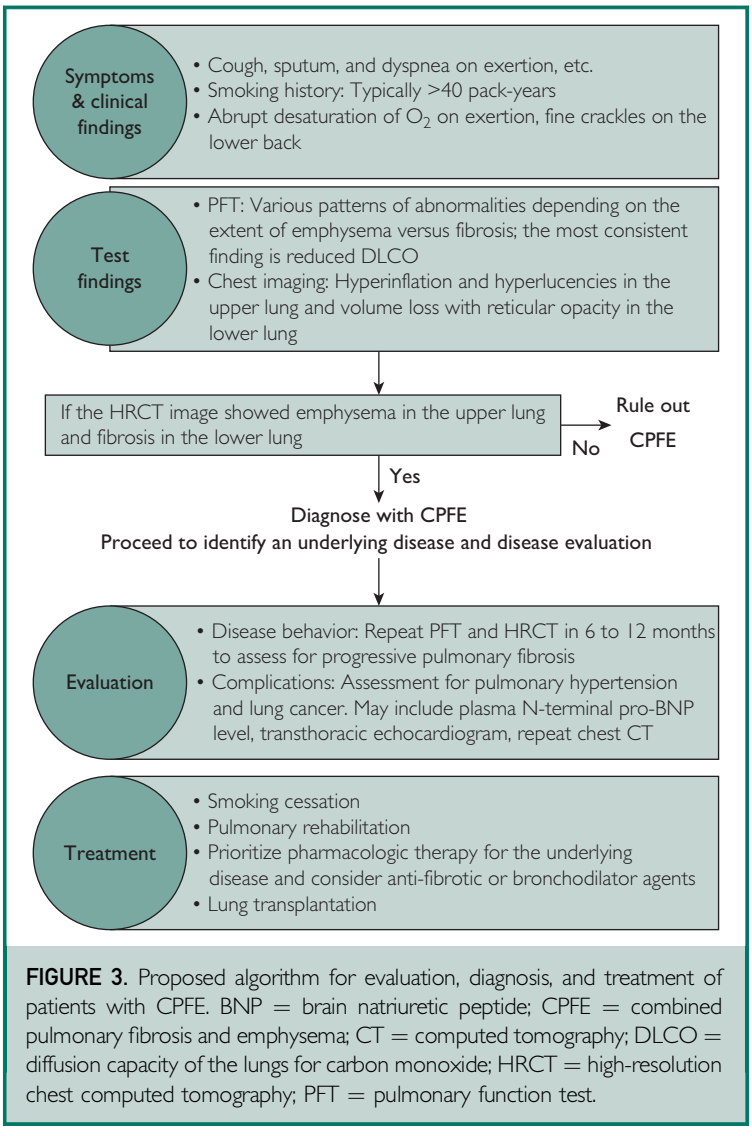


FIGURE 3. Proposed algorithm for evaluation, diagnosis, and treatment of patients with CPFE. BNP = brain natriuretic peptide; CPFE = combined pulmonary fibrosis and emphysema; CT = computed tomography; DLCO = diffusion capacity of the lungs for carbon monoxide; HRCT = high-resolution chest computed tomography; PFT = pulmonary function test.

was considered a variant of IPF.[47,48] Additionally, some studies excluded patients with underlying CTD, whereas others did not. Table 2 lists various fibrotic subtypes of ILDs associated with CPFE. In addition, the extent of emphysema and pulmonary fibrosis needed to qualify as CPFE has varied. For example, some studies defined the inclusion criteria as emphysema of more than 10% (percentage of lung parenchyma affected) and pulmonary fibrosis of more than 10% or more than 15%,[26,49,50] but others did not use cutoff values. In the recent Fleischner Society position paper on ILAs, 5% threshold of fibrosis was suggested and is supported by a recent study showing the prognostic significance associated with this finding in patients with CPFE.[3,51] On the other hand, the international committee proposed the research definition of CPFE for 2022 comprising of (1) greater than or equal to 5% of emphysema of total lung volume, and (2) lung fibrosis of any subtype (any amount).[8] It is important to examine the validity of this definition.

PROGNOSIS AND COMPLICATIONS

Median survival associated with CPFE has been reported to range from 0.9 to 8.5 years in various studies.[1,11,44,52,53] The clinical course is affected not only by the nature of the underlying ILD, but also by complications that can occur in patients with CPFE, including lung cancer, acute exacerbation, pulmonary infection, and pulmonary hypertension.[1] Prognosis in patients with CPFE is determined by the severity of disease as assessed by DLCO and disease extent (fibrosis and emphysema) on CT, age, and the occurrence of complications, including lung cancer, acute exacerbation, and pulmonary hypertension.[8] Comparison of prognosis associated with CPFE to that of IPF or emphysema alone has been difficult and has yielded conflicting results because one must control for the severity of disease in some manner.[8]

The incidence of lung cancer in patients with CPFE is several-fold higher than that in patients with either COPD or IPF alone.[1] Furthermore, 19% of lung cancers in patients with CPFE were small-cell lung cancer, the most aggressive pulmonary malignancy.[54] Regions of the lung affected by lung cancer in patients with CPFE tend to be similar to those involved with fibrosis (ie, lower lobes [50% to 60%] and subpleural areas [60%]).[54-56] In a systematic review of patients with CPFE and lung cancer, 43% of the patients were staged as unresectable and squamous cell carcinoma was the most common subtype (42%).[57] Another study reported mortality related to lung cancer in CPFE to be three times higher compared with lung cancer occurring in patients with IPF.[58] Moreover, administration of lung

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cancer treatment can be complicated in patients with CPFE because surgical, pharmacologic, and radiation therapies can trigger an acute exacerbation (AE), often resulting in death.[59]

Acute exacerbation is a significant source of morbidity and mortality in patients with CPFE and can be related to COPD or ILD.[60] Among patients with COPD, exacerbations are commonly caused by viral and bacterial infections.[61] Noninfectious causes include air pollution and cigarette smoking.[62] However, it is unclear whether the incidence of COPD exacerbation in patients with CPFE is the same as in patients with COPD. In one retrospective study on CPFE occurring in patients with IPF, AE-IPF seemed to occur at a lower rate among patients with CPFE compared with IPF alone, whereas AE-COPD occurred at a similar rate compared with AE-IPF.[60] Multiple studies have reported AE during the clinical course in 10% to 30% of patients with IPF or other fibrotic ILDs.[63,64] Acute exacerbation in patients with ILDs commonly lacks identifiable triggers such as infections, surgery or other invasive procedures, and drugs.

With regard to the current coronavirus disease 2019 (COVID-19) pandemic, COPD and smoking were found to increase both the risk of infection and its severity.[65,66] One meta-analysis showed an odds ratio of 5.69 in patients with COPD for severe COVID-19 infection.[67] Furthermore, several studies reported ILD as a risk factor for poor outcomes.[68,69] Among those studies, a multicenter case-control study showed an odds ratio of 3.2 in patients with ILD for COVID-19—related death, and 33% of patients with ILD had a coexisting obstructive lung disease.[69]

As already mentioned, pulmonary hypertension is often encountered in patients with CPFE and portends a worse prognosis. The marked decrease in DLCO observed in patients with CPFE is due to the loss of the normal lung parenchyma caused by emphysema and fibrosis, as well as the development of pulmonary hypertension. Up to half of the patients with CPFE develop precapillary pulmonary hypertension.[70] However, the presence of pulmonary hypertension may be overlooked in these patients; echocardiography is important for screening and monitoring this complication. For comparison, the prevalence of pulmonary hypertension was reported to be 39% in patients with COPD in a recent systematic review[71] and estimated to affect 35% to 50% of patients with IPF.[50,72]

TREATMENT

A clinical flow from presentation to treatment after diagnosis and evaluation is depicted in Figure 3. A fundamental step towards minimizing CPFE progression is smoking cessation for current smokers. Although there are no direct data pertaining specifically to CPFE, pulmonary rehabilitation measures may improve exertional dyspnea and quality of life. Supplemental oxygen therapy is indicated for those who manifest hypoxemia at rest or with exertion,[73] and appropriate vaccinations including pneumococcal, influenza, and COVID-19 should be provided. The goals of care such as advanced care planning should be discussed with patients and their families.

There has been no clinical trial of pharmacologic therapy in CPFE. Given the heterogeneity in underlying ILDs and complex interactions with the emphysematous component, decisions regarding the optimal treatment of CPFE can be difficult and must be individualized. Treatment for the underlying ILD depends in part on whether the nature of ILD is predominantly inflammatory (eg, sarcoidosis) or fibrotic (eg, IPF), warranting anti-inflammatory/immunosuppressive or antifibrotic therapy, respectively. In addition, avoidance of additional exposure is crucial for patients with identifiable exogenous causes, such as drug-induced ILDs and hypersensitivity pneumonitis. For those patients with CPFE who manifest progressive fibrosis, it may be appropriate to consider using the antifibrotic drug such as nintedanib, which was shown to slow the progression in a broad spectrum of fibrotic ILDs in the recent INBUILD (Efficacy and Safety of Nintedanib in Patients

With Progressive Fibrosing Interstitial Lung Disease [PF-ILD]) study.[74,75] Although there are no data on how best to assess response to therapy in patients with CPFE, quantitation of lung parenchyma involved with fibrosis and emphysema, gas exchange parameters (DLCO and oxygen saturation), and severity of respiratory symptoms may best serve as parameters to monitor.

Bronchodilator therapy is appropriate for patients manifesting features of reversible airflow obstruction meeting the criteria of COPD. However, there are no data and no reason to expect benefit from bronchodilator therapy for patients with CPFE who show no obstructive impairment. Similarly, the potential for benefit from use of systemic glucocorticoids in the management of disease exacerbations in CPFE is unknown.

A recent study reported improved exercise capacity (assessed with a 6-minute walk test) with inhaled treprostinil therapy for patients with pulmonary hypertension due to ILD; 25% of the study cohort had CPFE.[76] However, there was no significant improvement in patient-reported quality of life compared with the placebo group. Lung transplantation may be an option in select patients younger than 70 years without extrapulmonary contraindications and who are deemed appropriate candidates.

CONCLUSION

Combined pulmonary fibrosis and emphysema is associated with significant morbidity and mortality. It is often unrecognized because of counterbalancing effects of the two processes on both chest radiography and spirometry but is readily identifiable with HRCT and measurement of DLCO. It is clearly more common than previously recognized and occurs with a broader spectrum of interstitial lung diseases than was initially described. Combined pulmonary fibrosis and emphysema is associated with an increased risk for the development of lung cancer and pulmonary hypertension. Increased awareness among clinicians and studies of potential treatment modalities are needed in identifying optimal management strategies for patients with this syndrome.

POTENTIAL COMPETING INTERESTS

The authors have reported that they have no potential competing interests.

Abbreviations and Acronyms: **AE,** acute exacerbation; **COPD,** chronic obstructive pulmonary disease; **CPFE,** combined pulmonary fibrosis and emphysema; **CT,** computed tomography; **CTD,** connective tissue disease; **DLCO,** diffusing capacity for carbon monoxide; **FEV_1,** forced expiratory volume in one second; **HRCT,** high-resolution computed tomography; **ILA,** interstitial lung abnormalities; **ILD,** interstitial lung disease; **IPF,** idiopathic pulmonary fibrosis; **PFT,** pulmonary function test

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